GG-G-76E
May 7, 1980
SUPERSEDING
Fed. Spec. GG-G-76D
December 4, 1975

FEDERAL SPECIFICATION

GAGES, PRESSURE AND VACUUM, DIAL INDICATING (FOR AIR STEAM, OIL, WATER, AMMONIA, CHLORO-FLUORO HYDROCARBON GASES, AND COMPRESSED GASES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers dial gages which indicate pressure or vacuum on a graduated dial by means of a pointer, utilizing an elastic element (Bourdon tube or coil) and actuating linkage, as required, for measuring the pressure or vacuum.

1.2 Classification.

1.2.1 Classes, styles, sizes, types, and models. The gages covered by this specification shall be of the following classes, styles, sizes, types, and models as specified (see 6.2):

Class 1 - Pressure gage.

Style X - Single.
Style Y - Duplex (two-pressures).
Style Z - Differential.

Class 2 - Vacuum gage.
Class 3 - Compound (pressure-vacuum) gage.

Size 2 - 2-inch nominal diameter.
Size 2-1/2 - 2-1/2 inch nominal diameter.
Size 3-1/2 - 3-1/2 inch nominal diameter.
Size 4-1/2 - 4-1/2 inch nominal diameter.
Size 6 - 6-inch nominal diameter.
Size 8-1/2 - 8-1/2 inch nominal diameter.
Size 10 - 10-inch nominal diameter.
Size 12 - 12-inch nominal diameter.

Type I - Gage for air, steam, oil, and water.
Type II - Gage for ammonia.
Type III - Gage for chloro-fluoro hydrocarbon gases.
Type IV - Gage for compressed gases.

Model A - Air, argon, helium, hydrogen, nitrogen, and oxygen.
Model B - Acetylene, air, argon, helium, nitrogen, and oxygen.
Model C - Oxygen, nitrogen, and hydrogen.
Model D - Acetylene.
Model E - Air, Nitrogen, and oxygen.

FSC 6685

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. H28 - Screw-Thread Standards for Federal Services.
Fed. Std. 66 - Steel: Chemical Composition and Hardenability.
Fed. Std. 123 - Marking for Shipment (Civil agencies).
Fed. Std. 406 - Plastics: Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Instructions in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Washington, DC; Philadelphia; Atlanta, Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, and commercial item descriptions, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation-Packaging, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 20260-26T11:48Z
Check the source to verify that this is the current version before use.

American National Standards Institute (ANSI) Standard:

B40.1 - Gages, Pressure and Vacuum, Indicating Dial Type - Elastic Element.

(Application for copies should be addresed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Underwriter's Laboratories Inc.:

UL 404 - Gauges, Indicating Pressure, for Compressed Gas Service.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a gage for first article inspection and approval (see 4.2.1 and 6.3).

3.2 Standard commercial product. The gages shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Design. The gages shall be new and shall embody the design characteristics stated for the respective class, style, size, type, and model. All parts or subassembly containing parts subject to wear, breakage, or distortion shall be accessible for adjustment and repair. The dial-indicating gages shall be Bourdon tube (or coil), actuated for measurement and indication of the pressure and vacuum, as applicable, within the ranges specified in table I. When specified (see 6.2), the scale shall be in U.S., U.S. and ISO metric, or ISO metric.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

TABLE I. Scale range, graduations, and figure intervals.

Gage	Scale range		Smallest scale graduations (max.)		Figure intervals (max.)	
	Vacuum In. Hg	Pressure psi	Vacuum In. Hg	Pressure psi	Vacuum In. Hg	Pressure psi
Class 1 Style X, and Y, types 1 and IV; Models A, B, C, D, and E	--	0-15	--	1/2 or	--	3 or 5
	--	0-30	--	1/2	--	5
	--	0-60	--	1 or 2	--	5 or 10
	--	0-100	--	1 or 2	--	10
	--	0-200	--	2	--	20
	--	0-300	--	5	--	50
	--	0-400	--	10	--	50
	--	0-500	--	10	--	50 or 100
	--	0-600	--	10	--	50 or 100
	--	0-800	--	10	--	100
	--	0-1,000	--	10	--	100
	--	0-1,500	--	20	--	200
	--	0-2,000	--	25	--	200
	--	0-3,000	--	25	--	500
	--	0-4,000	--	50 or 100	--	500
	--	0-5,000	--	50	--	500 or 1,000
	--	0-6,000	--	40	--	600 or 1,000
	--	0-10,000	--	100	--	1,000
	--	0-15,000	--	200	--	--
	--	0-20,000	--	200	--	--
Class 1, style Z type I	--	10-0-10	--	1/2	--	5
	--	15-0-15	--	1/2	--	5
	--	30-0-30	--	1	--	10
	--	50-0-50	--	1	--	10
	--	10-0-10	--	0.2	--	2
	--	15-0-15	--	0.2	--	2
	--	30-0-30	--	0.5	--	3
	--	50-0-50	--	0.5	--	5
	--	80-0-80	--	2	--	20
	--	100-0-100	--	2	--	20
	--	150-0-150	--	5	--	50
	--	200-0-200	--	5	--	50
	--	300-0-300	--	5	--	100
	--	400-0-400	--	10	--	100
	--	500-0-500	--	10	--	100
	--	600-0-600	--	10	--	100
Class 2, type I	0-30	--	1/2	--	5	
Class 3, type I	0-30	0-15	1	1	5	5
	0-30	0-30	1	1	10	10
	0-30	0-60	2	2	10	10
	0-30	0-100	2	2	30	10
	0-30	0-150	5	5	30	30
	0-30	0-200	5	5	30	25
	0-30	0-300	5	5	30	50

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

TABLE I. Scale range, graduations, and figure intervals - Continued.

Gage	Scale range		Smallest scale graduations (max.)		Figure intervals (max.)	
	Vacuum In. Hg	Pressure psi	Vacuum In. Hg	Pressure psi	Vacuum In. Hg	Pressure psi
Class 3, type II	0-30	0-150	5	5	30	25
	0-30	0-300	5	5	30	25
	0-30	0-60	2	1	10	10
	0-30	0-100	2	1	30	10
Class 3, type III	0-30	0-150	5	5	20	25
	0-30	0-300	5	5	30	25
Class 1 style X, type III	--	0-300	--	5	--	50
	--	0-400	--	10	--	50

3.3.1 Class 1. The class 1 gages shall be pressure-indicating gages involving the following styles:

3.3.1.1 Style X. The style X pressure gage shall incorporate a single Bourdon tube (or coil), for standard applications of pressure indication.

3.3.1.2 Style Y. The style Y pressure gage shall be a duplex pressure gage incorporating two Bourdon tubes (or coils), with actuating mechanisms, two inlet connections, and two pointers for indicating two separate and independent pressures on the single dial.

3.3.1.3 Style Z. The style Z pressure gage shall be a differential pressure gage incorporating two opposite action Bourdon tubes (or coils), two inlet connections, and a single indicator pointer. The gage shall indicate the differential pressure between two independent pressure sources.

3.3.2 Class 2. The class 2 gage shall be a vacuum gage incorporating a single Bourdon tube (or coil), pointer, and inlet connection designed for vacuum indications.

3.3.3 Class 3.

3.3.4 Type I. The type I gages shall be pressure, vacuum, or compound for use with air, steam, oil, or water.

3.3.5 Type II. The type II gage shall be a compound gage for use with the medium ammonia.

3.3.6 Type III. The type III gage shall be a compound or pressure gage for use with the refrigerant medium specified (see 6.2). When specified (see 6.2), the pressure scales of type III gages shall be marked with temperature equivalents.

3.3.7 Type IV. The type IV gages shall be a pressure gage for use with the compressed gases specified in 1.2.1, stored in cylinders, or for use with cutting and welding equipment, in gaging pressures of compressed gases charged to 2,500 pounds per square inch (psi), at 70° Fahrenheit (F). With back-connected model A gages, a device shall be provided capable of severely limiting the flow of gas to the Bourdon tube (or coil), should an explosion occur. Excluding the model D gage, all type IV gages shall be free of all traces or organic matter which might otherwise cause an explosion hazard, and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

the gage dial shall be marked for oxygen use.

3.3.8 Size. The nominal size of the gage shall be the inside diameter of the case in inches, measured at the face of the dial.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4.1 Dissimilar metals. Contact between dissimilar metals, especially brass, copper, or steel in contact with aluminum or magnesium shall be avoided wherever possible.

3.5 Construction. The gage shall be complete so that when installed it can be used for the operations specified herein.

3.5.1 Identical items. All gages of the same classification, furnished under any specific contract, shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviation will be acceptable without prior written approval of the contracting officer.

3.5.2 Castings and forgings. All castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching and other defects that may affect the structural strength of the forgings involved. Strength and other essential physical and chemical properties of the castings and forgings shall be adequate to meet the performance requirements specified herein.

3.5.3 Welding, brazing, and soldering. Welding, brazing, and soldering shall be performed in accordance with the best commercial practices for the particular application. In no event shall such processes, including peening or plugging, be used on castings or forgings for reclaiming any parts of the gage without authorization of the procuring activity.

3.5.4 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in a manner that will prevent loss of tightness. When subject to removal or adjustment, such parts shall not be swaged, peened, staked, or otherwise permanently deformed.

3.5.5 Threads. All threaded parts shall be in the inch system and shall comply with Fed. Std. H28.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

3.5.6 Safety device. Each gage shall be furnished with safety blowout protection. Except for class 2 gages, blowout protection shall provide for equalization between ambient and case interior pressures at a maximum differential of 4 psi. The gage shall be so constructed that in the event of bursting of the elastic element, or its means for joining, due to an internal explosion or other application of excessive pressure, the gases shall escape in a direction away from the face of the gage. The face of the gage shall not burst. Special safety devices shall be provided when required (see 3.3.7, 3.7.2, and 6.2). The inlet orifice to the Bourdon tube of all gages shall be limited to a maximum of 0.00154 square inch area.

3.5.7 Interchangeability. All replaceable parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model can be replaced or adjusted without requiring modification.

3.5.8 Surfaces. External surfaces shall be smoothed and all sharp edges and corners removed except when sharp edges and corners are required by design or performance.

3.6 Performance and product characteristics. The gages shall embody the following performance and product characteristics:

3.6.1 Initial accuracy. Unless otherwise specified (see 6.2), the gages shall have the following initial accuracy at a temperature of 70° ±15°F. The error in pressure or vacuum indications shall not exceed ±1 percent of the full scale value in the middle half section of the scale and ±2 percent of the full scale value in the first and last one-fourth section of the scale.

3.6.2 Pressure whip requirement. The pressure gage shall be capable of withstanding without damage or change in accuracy of more than 1/2 percent of full scale value from the initial accuracy readings of 3.6.1, a total of 28,800 cycles of pressure, as shown in table II for the respective size pressure gage.

TABLE II. Range of pressures.

Dial size	Scale range	Cycle excursion percent of scale range
Inches	Psi	
Under 3-1/2	Up to and including 0-800	20 to 80
3-1/2 and over	Up to and including 0-800	20 to 80
4-1/2 and over	0-1000 up to and including 0-3000	20 to 80
4-1/2 and over	Exceeding 0-3000 and up to and including 0-5000	30 to 70
3-1/2 and under	Exceeding 0-800	40 to 60
4-1/2 and over	Exceeding 0-5000	40 to 60
3-1/2 and under	Up to and including 0-800	25 to 70

3.6.3 Excessive pressure requirement. Class 1 and 3 gages shall be capable of withstanding, without damage or change in accuracy, a pressure for a period of 1/2 hour of not less than 1.3 times the maximum scale reading for gages having a scale range up to and including 0 to 800 psi; 1.1 times the maximum scale reading for gages having a scale range from 0 to 1,000 psi up to and including 0 to 5,000 psi; and 1.0 times the maximum scale reading for gages having a scale range exceeding 0 to 5,000 psi.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

3.6.4 Tightness requirement. Class 1 and 3 gages shall be capable of retaining a pressure equal to not less than 80 percent of the full scale reading of the gage for a period of 1/2 hour without a drop in indicated pressure. The class 2 and 3 gages shall be capable of retaining a vacuum equal to not less than 80 percent of full scale reading for a period of 1/2 hour without a loss in indicated vacuum (see 4.6.4.1 and 4.6.4.2).

3.6.5 Storage temperatures. The gages and all components thereof shall be able to withstand storage without damage, in a temperature of -65°F for periods up to 72 hours, and a 24-hour cycle of temperature, starting with 10 hours at +90°F, followed by 5 hours of steady increase to +155°F, 4 hours at +155°F, and 5 hours of steady decrease to +90°F. Following these extreme exposures and when stabilized at +75° ±15°F, the gages shall conform to the accuracy requirements set forth in 3.6.1 (see 4.6.5).

3.6.6 Vibration. When specified, gages intended for severe duty (see 6.2), such as shipboard duty (high frequency and amplitude vibrations) shall be subjected to type I environmental vibration test as specified in 4.6.7. No evidence of damage shall occur other than a shift in calibration. Loosening of the threaded fasteners, bezel rupture of parts, shall be cause for rejection.

3.6.7 Burst pressure. The gages, except for class 2 gages shall be tested for internal burst at a minimum of 1-1/2 times the highest gage dial pressure reading. Bursting of the internal components at or below the test pressure shall be cause for rejection. A test of the internal element shall be conducted to determine if the gage meets this requirement (see 4.6.8.1). Safety blowout protection shall be tested to determine if the case meets the requirements of 3.5.6 (see 4.6.8.2).

3.7 Components. The gages shall consist of the case, inlet connection(s), dial, pointer, gage adjustment, and Bourdon tube (or coil) mechanism.

3.7.1 Case. The gage case shall be phenol or ABS plastic or metal, as specified (see 6.2). Case depth shall be manufacturer's standard or as specified (see 6.2). The gage mounting (flush, surface, steam) required shall be as specified (see 6.2). When applicable, the body diameter for a panel opening and the bolt circle diameter shall be as specified (see 6.2). Gage cases shall be provided with means to prevent pressure buildup within the case. The gage case shall be fitted with not less than 1/16-inch shatterproof glass, plastic crystal, or one-piece plastic ring and window, clear and free from blemishes (see 4.6.6). The crystal shall be secured to the case with a snap-ring fixed in a recessed groove, a cam-lock bayonet type connection, a snap-over ring, or a threaded or hinged bezel. When threaded, the first two threads shall be turned down to the root diameter of the threads. The crystal shall be cushioned to reduce cracking and the case sealed to prevent the entrance of dust and moisture by a felt, cork, polyprophlene, or neoprene gasket.

3.7.1.1 Plastic case. The plastic case shall not warp, crack, or fade in color, be free from checks, and chipped or blistered surfaces, shall have no burrs or projecting ridges at the dividing line of the mold, shall be flame resistant, shall be flat black in color, and have a smooth surface (see 4.6.6).

3.7.1.2 Metal case. Metal cases shall be of the following materials and finish, as specified (see 6.2):

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:48Z
Check the source to verify that this is the current version before use.

(a) Brass, polished, and finished with black enamel (type I and III gages only).
(b) Steel zinc-coated or phosphate treated and finished with black enamel.
(c) Cast iron, zinc-coated, or phosphate treated and finished with black enamel.
(d) Aluminum alloy with anodized coating or chemical conversion coating, and finished with black enamel.
(e) Corrosion-resistant steel conforming to any of the ANSI 300 series steels of Fed. Std. 66. The finish shall be a No. 4 standard commercial polish or better.

3.7.2 Inlet connection. The gages shall be fitted with a single inlet connection, except class 1, styles Y and Z, which shall have two connections. The inlet connection(s) shall be 1/8-inch, 1/4-inch, or 1/2-inch NPT male connection as specified (see 6.2). Inlet conection(s) shall be located at the bottom or back of the gage case, as specified (see 6.2). The effective thread length of the connection(s) shall be as specified in Fed. Std. H28. The inlet connection(s) shall incorporate a wrench grip of not less than 1/8 inch for a 1/8-inch NPT fitting, 1/4 inch for a 1/4-inch NPT fitting, and 1/2-inch for the 1/2-inch NPT fittings. The inlet connection of each type IV, model A or B gage shall be provided with a baffle plug or restrictor designed to prevent an inrush of high pressure gas which may result in gage damage. Gages having ranges of 20,000 psi and above shall be furnished with 1/4-inch high pressure tubing connection unless otherwise specified (see 6.2).

3.7.3 Dials. The dial graduations and figures shall be black on a white background, white on a black background, or luminous yellow on black background. The dial scale shall cover an arc of not less than 270°. The graduated dial scale arc length and dial figure height shall be in accordance with table III. The dial shall be of plastic material, aluminum, or steel protected against corrosion. The gages shall incorporate the scale ranges, graduations, and figure intervals as specified (see 6.2), from the tabulated values in table I. The gages involving a divided scale shall have the vacuum and pressure scales clearly identified. The dial of class 1, style Z gage shall be marked to indicate the inlet connection for the greater pressure source. The type II gage shall have the word "Ammonia" permanently and plainly inscribed on the dial face. The type III gage shall have the type refrigerant permanently and plainly inscribed on the dial face. When required, a retarded scale shall be furnished with scale range, graduations, and figure intervals as specified (see 6.2). When specified (see 6.2), the dial of type IV, model A gage shall be equipped with a scale to indicate the volume of gas remaining in the cylinder at 70°F. If used, the scale shall be easily readable and shall be applied to the dial in accordance with the manufacturer's standard practice. Graduations on the dial of model D gages shall be to mid-dial in 1/2 or 1-pound (lb) scale divisions from 0 to 15 psi. The remainder of the dial may or may not be graduated, but shall be banded or indicated by a broad red line. The words "USE NO OIL" shall appear very conspicuously on the dial of all type IV gages. When the gage is restricted to a particular gas or type of gas, the name of the gas or type of gas shall be inscribed on the dial face.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

TABLE III. Dial dimensions.

Gage dial nominal diameter	Minimum height of figures	Minimum length of scale arc
Inches	Inches	Inches
2	1/8	4
2-1/2	5/32	5
3-1/2	7/32	7
4-1/2	1/4	9-9/16
6	11/32	12-3/4
8-1/2	1/2	18-3/8
10	9/16	21
12	5/8	25

3.7.4 Pointer. The indicating pointer shall be shaped to permit accurate readings of the scale markings. The pointer shall not interfere with the gage dial at any point. The gages shall be fitted with a black pointer if the dial background is white or fitted with a white pointer if the dial background is black. In addition, the class 1, style Y, type I gage shall be fitted with a red or green pointer. When specified (see 6.2), the gages, except class 1, style Z, type I, shall be fitted with any adjustable red marking pointer which can be secured at any point on the dial scale.

3.7.5 Gage adjustment. The gages shall be provided with an external zero adjustment. Removal of the gage crystal is permitted for adjustment of the gage. Micrometer-type pointer is an acceptable zero adjustment. The range of the zero adjustment on either side from the center of the zero index shall be no less than 3 percent of the full scale value.

3.7.6 Bourdon-tube mechanism. The Bourdon-tube mechanism shall consist principally of a Bourdon tube, socket, and actuating linkage movement embodying the following characteristics:

3.7.6.1 Bourdon tube. All gages shall have Bourdon tubes (or coils), fabricated from alloys capable of withstanding without detrimental effect, the media specified (see 6.2). The Bourdon tubes (or coils) shall be seamless or welded, and secured to the socket and tip by a stress relieved, corrosion-resistant joint, not affected by the medium involved. When specified, an external bleeder shall be provided as an integral part of the Bourdon tube (see 6.2).

3.7.6.1.1 Acetylene gages. To prevent a dangerous level of acetylides, the copper content of components in contact with acetylene shall not exceed 70 percent. The silver content of filler metals in brazing shall be limited to a maximum of 41 percent silver and 20 percent copper. Bourdon tubes or coils for high pressure gages (above 580 psi) shall not contain any copper alloys.

3.7.6.2 Socket and actuating linkage. The socket and actuating linkage shall be of corrosion-resistant material neutral to the effects of the medium specified. The socket shall incorporate the inlet connection specified in 3.7.2.

3.8 Standards compliance. Unless otherwise specified (see 6.2), gages shall conform to the requirements for grade A gages of ANSI B40.1, in addition to the requirments specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

3.9 Identification marking. The gages and where practicable, replaceable parts shall be marked for identification in accordance with standard commercial practice.

3.10 Technical publications. When specified in the contract, the contractor shall furnish instructions (as defined in 6.3).

3.11 Treatment and painting. Unless otherwise specified (see 3.7.1 and 6.2), the gages shall be treated and painted in accordance with the manufacturer's best standard practice.

3.12 Workmanship. The quality of workmanship shall meet the standards prevalent among manufacturers who normally produce equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the contractor is responsible to assure that materials and components used where manufactured, examined, and tested in accordance with the specified requirements of referenced subsidiary specifications and standards to the extent specified herein, or if none, in accordance with this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one gage when a first article sample is required (see 3.1 and 6.4). Unless otherwise specified (see 6.2), this inspection shall include the examination of 4.5 and the tests of 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection. The first article may be a standard production gage from the contractor's current inventory provided the gage meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining units to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on the sample gages selected in accordance with 4.4. This inspection shall include the examination of 4.5 and the tests of 4.6.1 and 4.6.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

4.3 Inspection lot. All gages of the same class, style, size, type, and model as applicable, offered to the Government at one time, shall be considered a lot for purposes of inspection. The sample unit shall be one complete gage.

4.4 Sampling. A random sample of gages shall be selected from each lot in accordance with MIL-STD-105.

4.4.1 Sampling for examination. Examination of the gages shall be based on inspection level II and an Acceptable Quality Level (AQL) of 4.0 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.2 Sampling for tests. Tests of the gages shall be based on inspection level S-2 and an AQL of 4.0 percent defective.

4.5 Examination. The samples of gages selected in 4.4.1 shall be examined visually in accordance with the classification of defects listed in table IV, and shall include all defects classified in accordance with MIL-STD-105.

TABLE IV. Classification of defects.

Major:	
101	Gage inoperative (see 3.5).
102	Defective castings and forgings affecting operation (see 3.5.2).
103	Defective welding, brazing, or soldering affecting operation (see 3.5.3).
104	Defective fastening devices affecting operation (see 3.5.4).
105	Safety devices not as specified (see 3.5.6).
106	Omission of a component part (see 3.7).
107	Gage cases not provided with pressure relief provisions (see 3.7.1 and 3.7.2).
108	Gage dimensions not as specified (see table III).
109	Gage crystals and bezels not as specified (see 3.7.1).
110	Gage inlet connections not as specified (see 3.7.2).
111	Dial size, figures, scale range, scale divisions, and markings not as specified (see 3.7.3).
112	Points not as specified (see 3.7.4).
113	Gage zero adjustment not as specified (see 3.7.5).
114	Bourdon tubes or coils not as specified (see 3.7.6 through 3.7.6.2).
Minor:	
201	Threads not in accordance with Fed. STd. H28 (use thread gage) (see 3.5.5).
202	Surfaces not smooth (see 3.5.8).
203	Finishes for metal gage cases not as specified (see 3.7.1.2).
204	Identification marking not in accordance with standard commercial practice (see 3.9).
205	Sheet of instructions omitted or incomplete (see 3.10).
206	Workmanship not as specified (see 3.12).

4.6 Test methods. The accuracy test in 4.6.1 shall be performed prior to any other test.

4.6.1 Accuracy tests. The initial accuracy test shall be performed in an ambient room temperature of 75° ±15°F. The gages shall be tested at every numerical marking to determine compliance with 3.6.1. The gage tester shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

have an accuracy within 1/4 of 1 percent. A mercury liquid manometer is recommended for testing vacuum gages, in addition to pressure gages having pressure ranges not exceeding 30 psi.

4.6.2 Pressure whip test. With an applicable Bourdon gage whip tester used as indicated in figure 1, or in an equivalent arrangement, subject the pressure gage to 28,800 cycles of pressure as shown in table II On conclusion of the test operation, the gage shall be examined for damage and tested for accuracy in accordance with 4.6.1 to determine compliance with 3.6.2.

4.6.3 Excessive pressure test. With an applicable pressure gage testing unit, subject the pressure gage to the applicable pressure specified in 3.6.3 for a period of 1/2 hour. On conclusion of the test the gage shall be examined for damage and tested for accuracy in accordance with 4.6.1 to determine compliance with 3.6.3.

4.6.4 Tightness test.

4.6.4.1 For class 1 and 3 gages. At a stabilized temperature, the gages shall be connected to an applicable pressure source with a leak-proof shutoff valve installed at the inlet connection. A pressure equal to not less than 80 percent of the full pressure scale reading shall be applied and entrapped in the gage at the shutoff valve for a period of 1/2 hour. The gage shall be observed for any pressure drop to determine compliance with 3.6.4.

4.6.4.2 For class 2 and 3 gages. At a stabilized temperature the gage shall be connected to an applicable vacuum with a leak-proof shutoff valve installed at the inlet connection. A vacuum equal to not less than 80 percent of the full vacuum scale reading shall be applied to the gage and entrapped in the gage at the shutoff valve for a period of 1/2 hour. The gage shall be observed for any vacuum loss to determine compliance with 3.6.4.

4.6.5 Storage tests.

4.6.5.1 Low temperature test. The gages shall be placed in an ambient air temperature of -60° , to -65°F for a period of 72 hours. At the end of this time, the gages shall be removed and allowed to stabilize at an air temperature of +75° ±15°F. The gages shall then be inspected for evidence of damage and tested for accuracy in accordance with 4.6.1 (see 3.6.5).

4.6.5.2 High temperature test. The gages shall be submitted to a 24-hour cycle of temperature starting with 10 hours at +90° ±5°F, followed by 5 hours of steady increase to a range between +150° to +155°F; 4 hours to a range between +150° to +155°F; and 5 hours of steady decrease to a range of +90° ±5°F. The gages shall then be inspected for evidence of damage and tested for accuracy in accordance with 4.6.1 (see 3.6.5).

4.6.6 Plastic crystal and case test. The plastic crystal and case shall be tested as specified in Fed. Std. 406 to determine compliance with 3.7.1 and 3.7.1.1.

4.6.7 Vibration test. When specified in the contract that the gages will be utilized for severe duty, the gages shall be subjected to type I Environmental Vibration test of MIL-STD-167 to determine compliance with 3.6.6. The

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:48Z
Check the source to verify that this is the current version before use.

gage shall be tested when energized at high gage reading and when mounted in a similar manner as the gage will be mounted during service application as indicated on the contract (see 6.2).

4.6.8 Burst test. The Bourdon tube or coil and gage case shall be tested for resistance to burst, the blowout protection disk to release pressure buildup, and the gage case to resist breakage in compliance with 3.5.6 and 3.6.7.

4.6.8.1 Bourdon tube or coil. Those parts of the pressure gage that are in contact with the gas shall not leak when tested to a pressure corresponding to 1.5 times the maximum scale reading for a period of not less than 1 minute to determine compliance with 3.6.7.

4.6.8.2 Blowout protection and case strength. A pressure gage case shall be connected to a high pressure and the energy E = pV(p = 1.5 times maximum scale reading, but not less than 580 psi, and V = Volume of the Bourdon tube or coil) shall be suddenly released into the gage casing. All openings shall be closed in the case for the test, as it would be during a working application. The entry passage into the gage casing shall not be less than 0.197-inch diameter. The blowout disk shall release the pressure. The case and the gage face shall not burst. The gases shall escape in a direction away from the face of the gage to determine compliance with 3.5.6 and 3.6.7.

4.7 Preparation for delivery inspection. The packaging, packing, and marking of the gages shall be inspected to verify conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Packaging. Each gage with technical publications, when furnished, shall be packaged together in accordance with MIL-P-116, method 1A.

5.1.1.2 Intermediate packaging. Gages of like description, weighing up to 10 lb shall be intermediate packaged in fiberboard boxes conforming to PPP-B-636, class weather-resistant. The gages shall be cushioned to prevent movement within the boxes. The boxes shall be closed in accordance with the appendix to the box specification, method IV. Unless otherwise specified (see 6.2), the number of gages in each in each intermediate package shall not exceed 20 lb gross weight. Gages over 10 lb shall be packed directly into the shipping container.

5.1.2 Level C. Gages shall be packaged in a manner that will assure adequate protection against deterioration and damage during shipment. This level may conform to the contractor's commercial practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Gages shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, grade V2s. The contents shall be blocked and braced to prevent movement inside the container or damaging of the contents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. Gages shall be packed as specified for level A, except the boxes shall conform to PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. When specified (see 6.2), fiberboard boxes conforming to PPP-B-636, shall be grade V3c or V3s.

5.2.3 Level C. The gages shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules and National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract, shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military. In addition to any special marking required by the contract, the unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

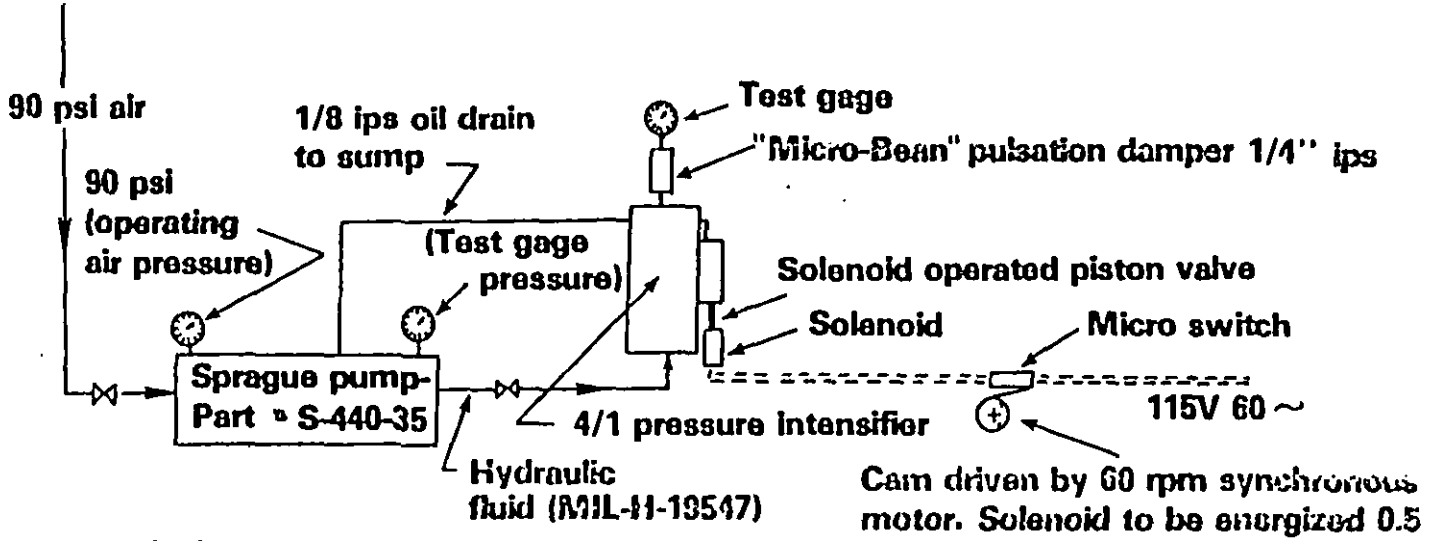
6. NOTES

6.1 Intended use. The gages covered by this specification are intended as applicable, for indicating pressures and vacuums. The type I gages shall be air, steam, oil and water; type II gages shall be for ammonia; type III gages shall be for chloro-fluoro hydrocarbon gases; and type IV gages shall be for compressed gases and mixtures listed in 1.2.1, in cylinders and distribution systems.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class, style, size, type, and model required (see 1.2.1).
(c) When first article is required for inspection and approval (see 3.1 4.2.1, and 6.4).
(d) Scale, (U.S., Dual U.S. and ISO metric, or ISO metric) (see 3.3) range graduations, and figure intervals required (see 3.7.3).
(e) Refrigerant to be used with type III gage, and whether the pressure scale marked with a temperature equivalent is required. (see 3.3.6).
(f) Special safety devices when required (see 3.5.6).
(g) Initial accuracy if different (see 3.6.1).
(h) Plastic or metal case required. Type of mounting, type of equipment mounted on, vibration frequency of equipment or mounting base (cycles per minute and magnitude of vibration (see 4.6.7), and when necessary, body diameter and bolt circle diameter (see 3.7.1).
(i) Gage case depth (see 3.7.1).
(j) Case material and finish (see 3.7.1.5).
(k) Inlet connection size required and location (see 3.7.2).
(l) Retarded scale range, graduations, figure intervals and a scale for type IV, model A gage to indicate volume of gas remaining in the cylinder at 70°F, when required (see 3.7.3).
(m) Adjustable marking pointer when required (see 3.7.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.



1. The installation of this device requires:

   A. A source of constant oil pressure which may be adjusted from 0 to 4,000 psi.

   B. A timing switch for the solenoid, 1/4 second on, 1/2 second off.

   C. A pressure snubber to soften the pressure shock to the gage.

NOTE: Insert gas may be used for gage having a range not exceeding 100 psi maximum.

FIGURE 1. Whip Test Arrangement

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

(n) Medium, gases, or mixtures on which gages are to be used, and bleeder requirements (see 1.2.1 and 3.7.6.1).
(o) Standards compliance, if different (see 3.8).
(p) Treatment and painting, if different (see 3.11).
(q) First article inspection, if different (see 4.2.1).
(r) Level of packaging and level of packing required (see 5.1 and 5.2).
(s) If intermediate package quantity is other than specified (see 5.1.1.2).
(t) When fiberboard boxes conforming to PPP-B-636, grade V3c or V3s are required for level B packing (see 5.2.2).

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article is required for Military procurements, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangement for examination, test, and approval of the first article.

6.5 Bourdon tubes. The term "Bourdon tube (or coil)" as used in this specification shall be interpreted to mean a tubular elastic sensing element having any of the various geometries currently offered in manufacturer's standard commercial products (see 3.3).

6.6 Safety recommendations. The maximum pressure at which a pressure gage is continuously operated shall not exceed 75 percent of full scale pressure. The gage selected should have a full scale pressure of approximately twice the intended operating pressure. This does not apply to retarded or suppressed scale gages. Contact supplier for recommendations.

MILITARY CUSTODIANS:

Army - ME
Navy - YD
Air Force - 99

Review activities:

Army - MI, AT, AR
Navy - SH, AS
DLA - GS

User activity:

Navy - MC

CIVIL AGENCY COORDINATING ACTIVITIES:

Interior - GES
GSA - FSS, PCD

Preparing activity:

Navy - YD

Project No. 6685-0581

U.S. GOVERNMENT PRINTING OFFICE : 1989 - 311-147/1136

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.